

Tutorial: Aggregate Expenditure Model - Numerical Exercises

Macroeconomics: ECON 101

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Introduction

The Aggregate Expenditure (AE) Model is a fundamental tool in macroeconomics for analyzing equilibrium output. It is given by:

$$AE = C + I + G + NX$$

where:

- C = Consumption
- I = Investment
- G = Government Spending
- NX = Net Exports (Exports - Imports)

The equilibrium GDP (Y^*) is found where $AE = Y$. The following exercises will help reinforce key concepts such as equilibrium output, the multiplier effect, fiscal policy, and external shocks.

Exercise 1: Finding Equilibrium GDP

Suppose the economy's aggregate expenditure function is:

$$AE = C + I + G + NX$$

where:

$$C = 200 + 0.8Y$$

$$I = 150$$

$$G = 250$$

$$NX = 50$$

Questions:

1. Find the equilibrium level of output Y^* (where $Y = AE$).
2. If investment increases to $I = 200$, what happens to equilibrium output?

Exercise 2: The Multiplier Effect

Consider an economy where:

- Marginal propensity to consume (MPC) = 0.75
- Autonomous spending (C, I, G, NX) = 500

Questions:

1. Compute the spending multiplier using the formula:

$$Multiplier = \frac{1}{1 - MPC}$$

2. If government spending increases by 100, by how much does equilibrium GDP change?

Exercise 3: Fiscal Policy and Aggregate Expenditure

Suppose an economy is in equilibrium at $Y = 4000$, with the following consumption function:

$$C = 300 + 0.6Y$$

Planned investment, government spending, and net exports sum to 900.

Questions:

1. Verify that the initial equilibrium is correct by solving for Y .
2. If the government increases spending by 200, find the new equilibrium GDP.
3. What is the total change in GDP?

Exercise 4: Taxes and the Aggregate Expenditure Model

An economy has the following consumption function:

$$C = 100 + 0.75(Y - T)$$

where $T = 200$ (lump sum taxes). Other components are:

$$I = 150$$

$$G = 250$$

$$NX = 50$$

Questions:

1. Find the equilibrium GDP (Y^*).
2. If taxes increase to $T = 250$, what happens to equilibrium output?

Exercise 5: Savings and the Paradox of Thrift

An economy is initially at equilibrium where:

$$C = 200 + 0.8Y$$

$$I = 100$$

$$G = 300$$

$$NX = 0$$

The government encourages saving, and people respond by reducing their MPC from 0.8 to 0.7.

Questions:

1. Find the initial equilibrium GDP.
2. Find the new equilibrium GDP after the change in MPC.
3. Explain the economic intuition behind the Paradox of Thrift.

Exercise 6: The Effect of an Export Boom

An economy has the following data:

$$C = 250 + 0.9Y$$

$$I = 300$$

$$G = 450$$

$$NX = 50$$

Suddenly, net exports increase to 150 due to higher global demand.

Questions:

1. Find the initial equilibrium GDP.
2. Find the new equilibrium GDP after the export boom.
3. What is the implied multiplier in this economy?

Exercise 7: Balanced Budget Multiplier

$$C = 0.8(Y - 500)$$

Suppose an economy has:

$$Y = C + I + G + NX$$

$$Y = 0.8(Y - 500) + 1000$$

$$MPC = 0.8$$

$$C = 0.8(Y - 400)$$

$$Y = 0.8(Y - 400) + 400 + 500$$

$$0.2Y = -400 + 1000$$

- Initial government spending $G = 400$

$$0.2Y = -320 + 900$$

- Initial taxes $T = 400$

$$0.2Y = 580$$

$$Y^* = 2900$$

$$0.2Y = 600$$

- Investment and net exports sum to 500

$$Y^* = 3000$$

The government decides to increase both spending and taxes by 100.

Questions:

$$\text{Multiplier}_{\text{tax}} = -MPC / (1 - MPC)$$

$$\text{multiplier} = 1 / 0.2 = 5$$

based on this multiplier equilibrium GDP should increase by 500

$$= -0.8 / 0.2 = -4$$

1. Find the initial equilibrium GDP.
2. What is the new equilibrium GDP after the fiscal policy change?
3. Explain the balanced budget multiplier and verify if it equals 1.

$$\text{Net effect} = 500 - 400 = 100$$

Exercise 8: A Negative Demand Shock

Consider an economy with the following:

$$C = 300 + 0.75Y$$

$$I = 250$$

$$G = 400$$

$$NX = 50$$

A financial crisis causes investment to drop by 100.

Questions:

1. Find the initial equilibrium GDP.
2. Find the new equilibrium GDP after the investment shock.
3. How does the size of the multiplier influence the total GDP contraction?